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Que. $I = \begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$

$I_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ $I_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$= \begin{bmatrix} -10 & 0 & 30 \\ -40 & 0 & 80 \\ -30 & 0 & 50 \end{bmatrix}$ $= \begin{bmatrix} -10 & -40 & -30 \\ 0 & 0 & 0 \\ 30 & 80 & 50 \end{bmatrix}$

$I_x = 20+40+20 = 80$ $I_y = 20+40+20 = 80$

$= 160 - 80$ $= 160 - 80$

$= 80$ $= 80$

$R = \det(m) - R\,(\text{trace}(m))^2$

$\det(m) = \begin{bmatrix} (80)^2 & 80 \times 80 \\ 80 \times 80 & (80)^2 \end{bmatrix}$

$= \begin{bmatrix} 6400 & 6400 \\ 6400 & 6400 \end{bmatrix}$

$= 0$

Notes

$\text{trace}(m) = I_x + I_y = 80 + 80 = 160$

$R = 0 - (0.04)(160)^2$

$= 0 - 1024$ $= \boxed{-1024}$ Ans.